

Fine-Tuning DistilBERT for Three-Class Sentiment Classification of Amazon Electronics Reviews

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Abstract

The report introduces a deep learning approach for classifying sentiments into three classes using 50,000 Amazon Electronics reviews taken from Amazon Reviews 2023 (Amazon Reviews 2023, 2024; McAuley Lab, 2024). The purpose of the assignment is to classify customer review texts as belonging to Negative, Neutral and Positive sentiments based on their star ratings, i.e., map 1-2 stars to Negative, 3 stars to Neutral and 4-5 stars to Positive. In order to implement the required model, a pre-trained DistilBERT was selected and fine-tuned as the model architecture for sequence classification (Hugging Face, n.d.b), since the understanding of review sentiments requires consideration of context-specific language use, negations and contrasts that are beyond reach of standard sparse vector approaches. The pipeline included data fetching and preprocessing, exploratory data analysis, splitting the dataset into train-validation-test sets, tokenization, class weighting in the loss function, fine-tuning and evaluation using accuracy, macro-F1 score, precision, recall, individual class scores and a confusion matrix. In the end, the proposed pipeline achieved 85.78% test accuracy and

67.77% macro-F1 score after one epoch of fine-tuning. Particularly, F1-scores were equal to 0.9340 for the Positive sentiment and 0.3531 for the Neutral sentiment, which indicates that minority class instances were hard to classify for the model. Thus, it can be concluded that DistilBERT performs well for implementing a sentiment classification pipeline, although there are limitations associated with class imbalance and labeling quality.

1. Project Overview

This project aims to develop a small-scale but practical deep learning application for sentiment classification in an e-commerce setting. Online retailers such as Amazon receive large volumes of customer reviews, which makes it unrealistic for organisations to examine all feedback manually. Automated sentiment classification can support large-scale identification of customer satisfaction patterns, product quality issues and potential service risks.

The chosen problem involves three-class sentiment classification of Amazon Electronics reviews. The model receives a review text as input and predicts one of three sentiment categories: Negative, Neutral or Positive. This is a practical real-world business problem because customer reviews provide valuable signals about product reliability, user

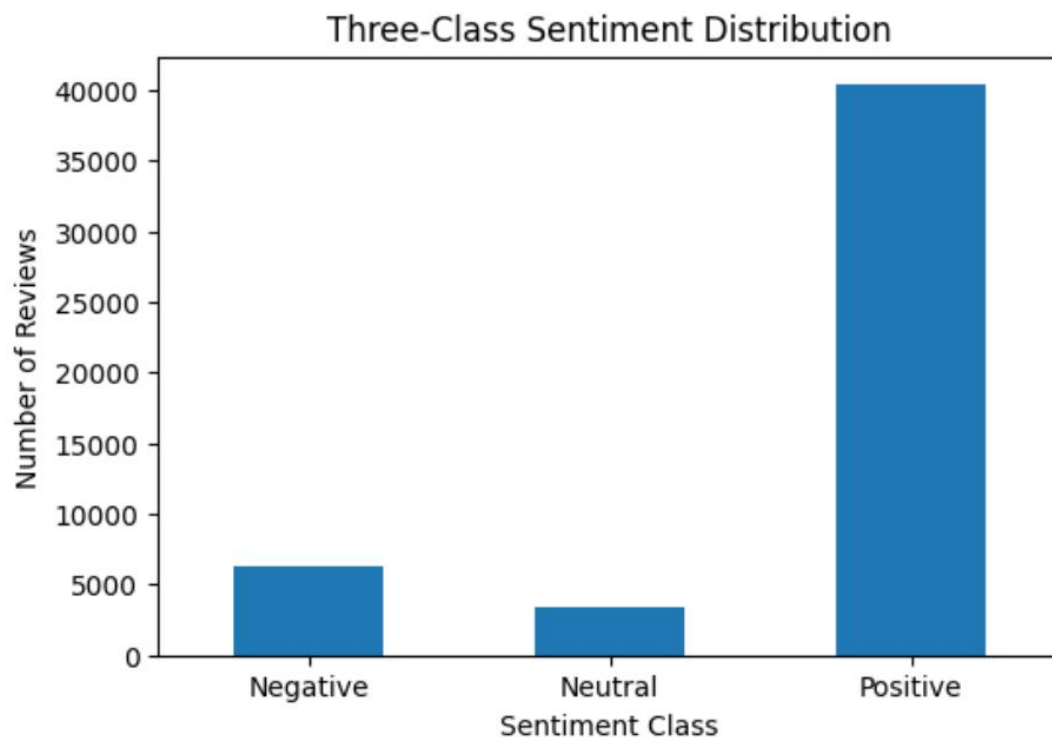
experience and purchase satisfaction. For example, a high number of negative reviews about battery life or product durability could help an organisation detect design flaws or prioritise customer support.

The solution was created as a reproducible Jupyter Notebook. The notebook streams 50,000 Amazon Electronics reviews from a remote source, maps star ratings to sentiment labels, fine-tunes a pre-trained DistilBERT model, and then evaluates the model using classification metrics and error analysis. The goal of this project is not to build a production-level review monitoring system, but to demonstrate a complete and critically evaluated deep learning pipeline for an applied natural language processing task.

2.Data Understanding

In this case, the used data set is an electronic subset of the Amazon Reviews 2023 data set. The features used include text review and star ratings. While there are more columns in the actual data set (title of the review, number of helpful votes and number of verified purchase status), since the project involves the classification of the sentiment of the text review, the star rating was used only to construct the sentiment label.

The selected data set included 50,000 samples. However, after label conversion, the resulting class distribution is highly imbalanced:



This distribution matters because it directly affects both model training and evaluation. Positive reviews make up the majority of the dataset, while Neutral reviews are the smallest class. As a result, relying on accuracy alone can give an overly optimistic view of model performance. A model that performs very well on Positive reviews but poorly on Neutral ones can still achieve high overall accuracy simply because most samples belong to the Positive class.

The way labels are constructed also introduces a limitation. Star ratings are used as proxy labels for textual sentiment, but they do not represent

perfect ground truth. A 3-star review may express mixed sentiment rather than clearly neutral language, while a 5-star review may be very short and lack strong positive wording. In the same way, a low rating might reflect issues with delivery or service rather than the product itself. This means the model is trained on weak labels derived from ratings, rather than on manually verified sentiment annotations.

3.Data Preparation and Setup

The notebook was designed to be self-contained and reproducible. Instead of loading data from local files, it streams the Amazon Electronics reviews dataset from a remote URL. This improves reproducibility, as the marker can run the notebook without needing to manually download the dataset.

Only minimal text cleaning was applied, including converting inputs to strings, removing simple HTML patterns and normalising whitespace. More extensive preprocessing, such as removing punctuation or lower-level linguistic features, was intentionally avoided because DistilBERT uses its own tokenizer and is designed to handle natural language text directly. Over-cleaning could remove important sentiment signals, such as negation or punctuation.

The dataset was split using a stratified 70/15/15 train-validation-test split:

Split	Size	Negative	Neutral	Positive
Train	34,993	12.52%	6.82%	80.65%
Validation	7,499	12.52%	6.83%	80.65%
Test	7,499	12.52%	6.83%	80.65%

Stratification was necessary due to the strong class imbalance, as it ensured that all three splits maintained approximately the same class proportions as the original dataset. The validation set was used to monitor training performance, while the test set was reserved for final evaluation.

The maximum sequence length was set to 128 tokens, reflecting a balance between retaining sufficient textual information and controlling computational cost. While longer sequences can capture more context from reviews, they also increase memory usage and runtime. The model was fine-tuned for one epoch to keep the notebook compatible with “Run All” execution in the assessment environment. In a more comprehensive experiment, training for around 3 – 5 epochs with early stopping based on validation macro-F1 would be more appropriate.

4. Model Selection and Traditional Machine Learning Comparison

Firstly, a traditional machine learning approach to this problem would typically apply TF-IDF vectorisation followed by a classifier such as Logistic Regression or Support Vector Machine. This approach has clear advantages: it is computationally efficient, relatively interpretable, and often performs well in simpler sentiment classification settings where individual words like “excellent”, “broken” or “poor” act as strong indicators.

However, TF-IDF-based methods have notable limitations for this task. They represent text mainly as sparse word-frequency features and struggle to capture word order, negation and contextual meaning. For instance, phrases like “not bad” , “good at first but failed later” , or “ average but acceptable for the price ” require context-aware interpretation. While a bag-of-words model may identify sentiment-related terms, it often fails to understand how those terms interact within a sentence.

Deep learning is therefore appropriate in this project because Amazon reviews are unstructured text data, and Transformer-based language models are specifically designed to learn contextual representations

from tokenised text (Vaswani et al., 2017; Devlin et al., 2019). It shifts the modelling process away from manually designed features towards learned representations. As highlighted in the course material, this is a key difference between traditional machine learning and deep learning: traditional approaches rely heavily on human-crafted features, whereas deep learning learns representations automatically and is particularly effective for unstructured data such as text, images and audio.

Several deep learning alternatives were considered. A model using trainable embeddings with a BiLSTM layer would be lighter and easier to train, and LSTMs are well-suited to sentiment analysis because they process sequences and can capture dependencies between words. However, such a model would learn word representations only from the 50,000-review dataset. A full BERT model could offer stronger contextual representations, but at the cost of higher memory usage and longer runtime. DistilBERT was therefore chosen as a practical compromise, as it provides pre-trained contextual representations while being smaller and faster than BERT (Sanh et al., 2019).

DistilBERT is built on the Transformer architecture. Unlike RNNs and LSTMs, Transformers process the entire sequence using self-attention, allowing each token to directly attend to all other tokens in the input.

This makes them more effective at capturing long-range dependencies and contextual relationships. In addition, Transformers can process full sequences in parallel and use self-attention to model interactions between tokens.

5.Implementation, Results and Evaluation

The implemented model used distilbert-base-uncased with a sequence classification head for three output classes. The tokenizer transformed raw review text into token IDs and attention masks, following the standard Hugging Face Transformers workflow (Hugging Face, n.d.a).The model was then fine-tuned using supervised labels derived from star ratings.

The main training configuration was:

Parameter	Value
Model	distilbert-base-uncased
Task	Three-class sequence classification
Max sequence length	128
Batch size	16
Learning rate	2e-5
Epochs	1
Optimiser	AdamW
Loss	Class-weighted cross-entropy

Class-weighted cross-entropy was used due to the strong class imbalance in the dataset. Without applying class weights, the model could overemphasise the majority Positive class. AdamW was chosen because it is widely used for fine-tuning Transformer models. Unlike standard Adam with L2 regularisation, AdamW decouples weight decay from the adaptive gradient update, which leads to more stable regularisation in large neural networks. AdamW is also a modern optimiser commonly used in Transformer training.

The final test results were:

Test Loss: 0.6174

Test Accuracy: 0.8578

Test Macro-F1: 0.6777

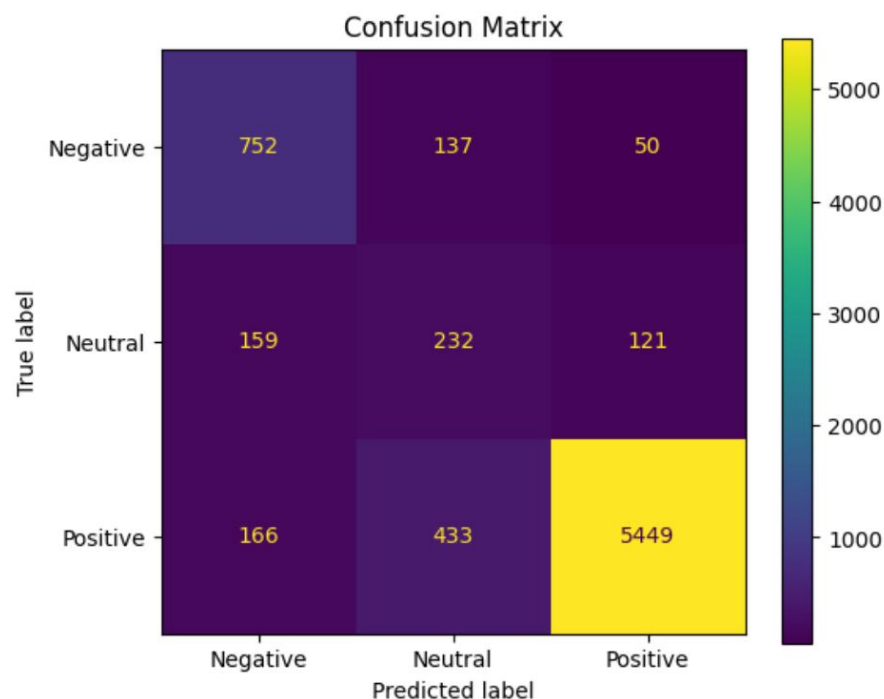
Classification Report:

	precision	recall	f1-score	support
Negative	0.6982	0.8009	0.7460	939
Neutral	0.2893	0.4531	0.3531	512
Positive	0.9696	0.9010	0.9340	6048
accuracy			0.8578	7499
macro avg	0.6524	0.7183	0.6777	7499
weighted avg	0.8891	0.8578	0.8708	7499

The model achieved high overall accuracy, but the macro-F1 score offers a more critical perspective on performance. The strong accuracy is partly explained by the fact that Positive reviews make up more than 80% of the dataset. The Positive class reached an F1-score of 0.9340, indicating

that the model was highly effective at identifying majority-class positive reviews. The Negative class also performed reasonably well, with an F1-score of 0.7460. In contrast, performance on the Neutral class was much weaker, with an F1-score of only 0.3531.

The confusion matrix further confirms this issue:



Only 232 out of 512 Neutral reviews were correctly classified. Many Neutral reviews were instead predicted as either Negative or Positive. This indicates that the Neutral category is not simply a midpoint between negative and positive sentiment. Rather, many 3-star reviews appear to contain mixed opinions, weak sentiment signals, or mismatches between the rating and the review text.

The gap between accuracy and macro-F1 is therefore one of the most important findings in the evaluation. While accuracy suggests strong overall performance, macro-F1 and the confusion matrix highlight uneven performance across classes. This reinforces the decision to report per-class precision, recall and F1-score, rather than relying solely on accuracy.

6.Critical Reflection, AI Use, Ethics and Implications

One of the main strengths of this implementation is that it delivers a complete and reproducible Transformer-based sentiment classification pipeline. The workflow covers remote dataset loading, label construction, exploratory data analysis, stratified splitting, model fine-tuning, class weighting, final test evaluation and error analysis. The choice of DistilBERT is also well suited to the task, as it leverages general language knowledge from pre-training and adapts it to the Amazon review domain through supervised fine-tuning.

However, the results also highlight several limitations. The weakest performance occurs in the Neutral class. This is partly due to class imbalance, as Neutral reviews make up only 6.82% of the dataset. At the same time, the issue is not purely numerical. The Neutral category is

inherently ambiguous: a 3-star review may include both positive and negative elements, making it difficult to classify as truly neutral. As a result, the low Neutral F1-score reflects both a modelling challenge and a labelling challenge.

Another limitation comes from using star ratings as sentiment labels. While ratings are convenient and scalable, they are imperfect proxies for textual sentiment. Some users assign high ratings with very brief or vague comments, while others give low ratings due to delivery, packaging or customer service issues rather than the product itself. Consequently, the model may learn patterns in user rating behaviour that do not fully align with the underlying sentiment expressed in the text.

The decision to train for only one epoch is also a limitation. This choice was made to keep the notebook reproducible within the runtime constraints of the assessment environment. In a more complete experiment, training for around 3 – 5 epochs with early stopping based on validation macro-F1 would be preferable. This could improve performance on minority classes, although any gains would need to be empirically validated.

AI assistance was used during development to explore model options, organise the notebook, debug code, improve documentation and refine the report structure. However, all final design decisions were reviewed and adjusted manually. These decisions included selecting 50,000 reviews, maintaining the three-class setup, choosing DistilBERT over a model trained from scratch, applying stratified splitting, using class-weighted loss and prioritising macro-F1 over accuracy. AI-generated suggestions were not adopted without critical evaluation.

The potential organisational impact of this solution is considerable. An e-commerce company could apply sentiment classification to monitor customer satisfaction, detect recurring product issues and prioritise support actions. For example, a sudden increase in Negative reviews for a product could signal quality problems. That said, such a system should not be used as the sole basis for decision-making. Misclassified reviews could lead to genuine complaints being overlooked, and missed signals of dissatisfaction could harm customer trust.

At the individual level, automated sentiment analysis could help ensure that customer feedback is addressed more quickly. However, there is a risk that ambiguous or mixed feedback is misinterpreted. Neutral reviews are particularly important because they often contain

constructive criticism. If they are consistently misclassified, organisations may lose valuable mid-range insights.

There are also ethical and environmental considerations. From a privacy standpoint, customer reviews may include personal information, so any production deployment should incorporate data minimisation and anonymisation. In terms of interpretability, DistilBERT is less transparent than traditional machine learning models. And deep learning models often function as black boxes and may require explainability techniques for responsible use. From an environmental perspective, fine-tuning DistilBERT requires more computational resources than approaches such as TF-IDF with Logistic Regression. Choosing DistilBERT instead of full BERT helps reduce this cost, but the trade-off between performance and computation remains.

7. Future Work

The first area for improvement is more systematic training. Future experiments should run for around 3 – 5 epochs and apply early stopping based on validation macro-F1. Further hyperparameter tuning could explore settings such as learning rate, batch size, maximum sequence length and class weighting strategies. This would help clarify

whether the weak performance on the Neutral class is mainly due to limited training, class imbalance or label ambiguity.

A second direction is to compare DistilBERT with other Transformer models, including RoBERTa, DeBERTa and domain-adapted BERT variants. These alternatives may offer stronger contextual representations, although they would also increase computational cost. Conducting such comparisons would make the trade-off between performance and efficiency more explicit.

A third opportunity lies in parameter-efficient fine-tuning. Methods such as adapters or LoRA (Low-Rank Adaptation) could reduce the number of trainable parameters while still adapting the model to the Amazon review domain. This is particularly relevant for organisations that need lower training costs and faster iteration cycles when deploying sentiment analysis systems.

A fourth direction is to move towards aspect-based sentiment analysis. The current model assigns a single sentiment label to an entire review, even though many Amazon reviews discuss multiple aspects. For example, a review might be positive about sound quality but negative about battery life. Future work could classify sentiment at the aspect

level rather than at the whole-review level, providing more actionable insights.

Finally, model explainability should be improved. Techniques such as attention visualisation, LIME, SHAP and integrated gradients could provide approximate diagnostic evidence about which tokens influence predictions. However, attention weights should not be treated as complete causal explanations of model behaviour. A human-in-the-loop approach could also be introduced, where low-confidence or high-risk predictions are reviewed manually.

8. Conclusion

This project developed a reproducible DistilBERT-based deep learning pipeline for three-class sentiment classification of Amazon Electronics reviews. The model achieved 85.78% test accuracy and 67.77% macro-F1, indicating strong overall performance but uneven results across classes. Positive reviews were classified very effectively, Negative reviews performed reasonably well, while Neutral reviews remained the primary weakness.

The findings demonstrate the value of pre-trained Transformer models for contextual sentiment classification, particularly when compared with

traditional TF-IDF-based approaches. At the same time, several important limitations are evident, including class imbalance, the use of weak rating-derived labels, poor performance on the Neutral class, lower interpretability and higher computational cost. Future work should focus on more extended and controlled fine-tuning, parameter-efficient adaptation methods, aspect-based sentiment analysis and improved explainability techniques.

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Appendix: AI Use Statement

AI assistance was used to support brainstorming, notebook structuring, code debugging, explanation of deep learning concepts and refinement of written expression. The final implementation decisions, interpretation of results and critical reflections were reviewed and adapted by the author. The submitted work reflects the author's own understanding of the dataset, modelling choices, evaluation strategy and limitations.